

CHE 494

Selected Topics in Chemical Engineering

1. Course number and name

CHE 494 Two-Dimensional (2D) Nanomaterials: Science, Processing, and Applications

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: Two, 75 min/wk. lectures; Engineering.

3. Instructor's or course coordinator's name

Sanjay Behura

4. Text book, title, author, publisher, year

2D Materials: Properties and Devices, P. Avouris, T. F. Heinz, and T. Low, Cambridge University Press, 2017.

a. Other supplemental materials

Raman Spectroscopy in Graphene Related Systems, A. Jorio, M. S. Dresselhaus, R. Saito, G., Dresselhaus; Selected Journal Articles from Nature, Science, Nano Letters, ACS Nano, JACS, PRL, PRB, and More.

5. Specific course information

a. Brief description of the content of the course (catalog description)

The goal of this course is to prepare students with foundational understanding of advanced two-dimensional (2D) nanomaterials including graphene, and the tools required to characterize such atomically-thin materials, and their potential applications.

b. Prerequisites or co-requisites

Consent of the instructor.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will:

- Understand what a 2D material is and how are they different from their bulk 3D counterparts.
- Learn how atomically-thin 2D materials and their quantum heterostructures are produced.
- Learn what are the unique and superlative properties of 2D materials and how to characterize them
- Understand the range of potential applications of the 2D materials
- Produce 2D materials in the lab

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.		X		
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	X			
(3)	an ability to communicate effectively with a range of audiences.			X	
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.				X
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.				X
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

Science of 2D Materials: Low-Dimensional Materials; Layered Materials; 2D Materials; 3D vs. 2D Materials; Modifications of Materials Properties: Doping in Bulk and 2D Materials; Fundamentals of Graphene, Hexagonal Boron Nitride, and Transition Metal Dichalcogenides; and Science of van der Waals 2D Heterostructures. (This part will focus on physics and chemistry of 2D materials, their unique structure and intriguing properties).

Processing of 2D Materials: Synthesis Techniques for Advanced 2D Materials (Graphene, Hexagonal Boron Nitride, and Transition Metal Dichalcogenides) and their van der Waals bound 2D Heterostructures. (This part will focus on both top-down and bottom-up strategies for the synthesis of 2D materials and understanding the ensuing growth mechanisms).

Characterization and Application of 2D Materials: Analysis of 2D Materials by Raman Spectroscopy, X-Ray Photoelectron Spectroscopy, Scanning Electron Microscopy, Transmission Electron Microscopy, and Scanning Probe Microscopy. Important Applications of 2D Materials including Nanoelectronics (Field Effect Transistors), Energy Conversion (Photovoltaic Solar Cells), and Energy Storage (Batteries). (This part will focus on basic working principles of advanced characterization tools, detailed analysis of 2D materials, and their potential applications).

2D Lab Work: Synthesis of Graphene on Metal Films via Chemical Vapor Deposition (CVD), Its Chemical Transfer, and Raman Spectroscopic and AFM Characterization.